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(57) Abstract :
FIELD ROVER FOR PLANT DISEASE MONITORING, FRUIT RIPENESS ANALYSIS, AND AUTOMATED SPRAYING ABSTRACT This project presents a smart agricultural rover designed for automated crop monitoring and targeted field intervention. The rover features an on board CMOS camera module to capture plant, leaf and fruit images by operating in the field. The Rover transmits these images to a wirelessly connected laptop where an At-based detection model computes disease classification and npeness of the fruit assessment in real time. When a disease is identified and the plant is tound infected the rover triggers a pesticide spraying mechanism that delivers a localized treatment. While the current prototype primarily focuses on imaging, analysis, and selectiye spraying, the system architecture is designed intentionally to support future extensions, optional modules such as robotic harvesting attachments or autonomous navigation capabilities using GPS guidance, sensor-driven movement, or visual-mapping techniques, though these features are not included in the present implementation.

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